MATH2601 – Higher Linear Algebra

Topic 6 – Eigenvalues and eigenvectors
Lecture 6.1 – Eigenvalues, eigenvectors, and diagonalisation

Lecturer: Dr Sean Gardiner – sean.gardiner@unsw.edu.au

Eigenvalues and eigenvectors

Definition. Given a vector space $(V, \mathbb{F})$ and $T \in \mathcal{L}(V, V)$, if $T(\mathbf{v}) = \lambda\mathbf{v}$ for some nonzero $\mathbf{v} \in V$ and some $\lambda \in \mathbb{F}$, we call λ an eigenvalue of T and $\mathbf{v}$ an eigenvector of T (also called a λ-eigenvector).

Definition. The set of all eigenvalues of $T \in \mathcal{L}(V, V)$ is called the spectrum of T and is written $\sigma = \sigma_T = \{\lambda \in \mathbb{F} : T(\mathbf{v}) = \lambda\mathbf{v}\}$.

The spectrum is sometimes written as a multiset, when certain eigenvalues have algebraic multiplicity (defined later on slide 8).

Definition. For $T \in \mathcal{L}(V, V)$ and a fixed $\lambda \in \mathbb{F}$, the set of all vector solutions to $T(\mathbf{v}) = \lambda\mathbf{v}$ is called the λ-eigenspace of T and is written $E_\lambda = E_\lambda(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \lambda\mathbf{v}\}$. So the λ-eigenspace is the set of all λ-eigenvectors of T, along with $\mathbf{0}$. Note that by definition, eigenvectors cannot be $\mathbf{0}$, but $\mathbf{0}$ is an element of every eigenspace.

As usual, we define eigenvalues and eigenvectors of any matrix $A \in M_{n,n}(\mathbb{F})$ by associating it with the mapping $T : \mathbb{F}^n \to \mathbb{F}^n$ given by $T(\mathbf{v}) = A\mathbf{v}$.

Example. For the matrix $\begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$, we find 4 is an eigenvalue with corresponding 4-eigenvector $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ since $\begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \end{pmatrix} = 4\begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

Linearly independent eigenvectors

Lemma. If $T \in \mathcal{L}(V, V)$ and $\lambda_1, \lambda_2, \ldots, \lambda_n$ are distinct eigenvalues of T, then any set $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ where each $\mathbf{v}_i$ is a λ_i-eigenvector is a linearly independent set.

Proof. Suppose by way of contradiction that $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly dependent, so $\mu_1\mathbf{v}_1 + \cdots + \mu_n\mathbf{v}_n = \mathbf{0}$ has nontrivial solutions for $\mu_i \in \mathbb{F}$. Over all such nontrivial solutions, choose that which has the fewest nonzero μ_i values and reorder the set of vectors so that the nonzero coefficients come first. Supposing there are k such nonzero coefficients, we have $\sum_{i=1}^{k} \mu_i\mathbf{v}_i = \mathbf{0}$, and so $T(\sum_{i=1}^{k} \mu_i\mathbf{v}_i) = \sum_{i=1}^{k} \mu_i T(\mathbf{v}_i) = \sum_{i=1}^{k} \mu_i\lambda_i\mathbf{v}_i = \mathbf{0}$.

On the other hand, multiplying the original linear dependence equation through by λ_k gives $\sum_{i=1}^{k} \mu_i\lambda_k\mathbf{v}_i = \mathbf{0}$, and the difference between this and the previous result is $\sum_{i=1}^{k} \mu_i(\lambda_i - \lambda_k)\mathbf{v}_i = \sum_{i=1}^{k-1} \mu_i(\lambda_i - \lambda_k)\mathbf{v}_i = \mathbf{0}$. Since the λ_i are distinct, this equation gives a nontrivial solution to the linear dependence equation with fewer nonzero μ_i terms than in the assumption. This implies $\{\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}\}$ is linearly independent, which contradicts the assumption that k was minimal.

Properties of eigenspaces

Lemma. Given $T \in \mathcal{L}(V, V)$, for any eigenvalue λ of T, we have $E_\lambda \leq V$.

Proof. Clearly $\mathbf{0} \in E_\lambda$ since $T() = \mathbf{0} = \lambda\mathbf{0}$ considering T is linear, so E_λ is nonempty. For all $\mathbf{u}, \mathbf{v} \in E_\lambda$ and $\mu \in \mathbb{F}$, we have
$T(\mathbf{u} + \mu\mathbf{v}) = T(\mathbf{u}) + \mu T(\mathbf{v}) = \lambda\mathbf{u} + \mu\lambda\mathbf{v} = \lambda(\mathbf{u} + \mu\mathbf{v})$, so $\mathbf{u} + \mu\mathbf{v} \in E_\lambda$.
Thus $E_\lambda \leq V$ by the subspace lemma.

Lemma. Given $T \in \mathcal{L}(V, V)$, for any eigenvalue λ of T, any subspace of E_λ is T-invariant.

Proof. Suppose $W \leq E_\lambda$. Then for all $\mathbf{v} \in W$, we have $T(\mathbf{v}) = \lambda\mathbf{v}$, which must be an element of W since subspaces are closed under scalar multiplication.

Lemma. Given $T \in \mathcal{L}(V, V)$, if λ and μ are distinct eigenvalues of T, their eigenspaces have trivial intersection, that is $E_\lambda \cap E_\mu = \{\mathbf{0}\}$.

Proof. Let $\mathbf{v} \in E_\lambda \cap E_\mu$. Then $T(\mathbf{v}) = \lambda\mathbf{v} = \mu\mathbf{v}$ for some distinct λ, μ. So $(\lambda - \mu)\mathbf{v} = \mathbf{0}$, and since $\lambda \neq \mu$, we must have $\mathbf{v} = \mathbf{0}$.

Characteristic polynomial

Lemma. Given $T \in \mathcal{L}(V, V)$, if λ is an eigenvalue of T, then its eigenspace is $E_\lambda = \ker(T - \lambda\,\mathrm{id})$. Likewise, if λ is an eigenvalue of $A \in M_{n,n}(\mathbb{F})$, then its eigenspace is $E_\lambda = \ker(A - \lambda I)$.

Proof. By definition, $\mathbf{v} \in E_\lambda$ implies $T(\mathbf{v}) = \lambda\mathbf{v}$, so $(T - \lambda\,\mathrm{id})(\mathbf{v}) = \mathbf{0}$. Thus $\mathbf{v} \in \ker(T - \lambda\,\mathrm{id})$.

Definition. Given $A \in M_{n,n}(\mathbb{F})$, the characteristic polynomial of A is given by $p_A(t) = \det(tI - A)$. The characteristic polynomial of a vector space homomorphism $T \in \mathcal{L}(V, V)$ is given by $p_T(t) = p_A(t)$ where $A \in M_{n,n}(\mathbb{F})$ for $n = \dim(V)$ is the matrix satisfying $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in V$.

(The characteristic polynomial is sometimes defined as $\det(A - tI)$; since we usually only care about its roots, the difference in sign is negligible.)

Lemma. Given $A \in M_{n,n}(\mathbb{F})$, the zeros of $p_A(t)$ are exactly the eigenvalues of A. So the spectrum of A is $\sigma_A = \{\lambda \in \mathbb{F} : p_A(\lambda) = 0\}$, and $|\sigma_A| \leq n$.

Proof. If λ is an eigenvalue of A, then $A\mathbf{v} = \lambda\mathbf{v} \iff (\lambda I - A)\mathbf{v} = \mathbf{0}$ for some $\mathbf{v} \neq \mathbf{0}$, so $\lambda I - A$ is not invertible, meaning $p_A(\lambda) = \det(\lambda I - A) = 0$. The same proof in reverse shows any zero of $p_A(t)$ is an eigenvalue of A.

Corollary. Given $T \in \mathcal{L}(V, V)$ with $\dim(V) = n$, we have λ is an eigenvalue of T if and only if $\mathrm{nullity}(T - \lambda\,\mathrm{id}) > 0$, if and only if $\mathrm{rank}(T - \lambda\,\mathrm{id}) < n$.

Example 1

Example. Find the eigenvalues and corresponding eigenspaces of the matrix $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & -1 & 3 \end{pmatrix}$.

Solution. The characteristic polynomial is

$$p_A(t) = \det(tI - A) = \begin{vmatrix} t-1 & 0 & 0 \\ 0 & t-1 & -1 \\ -1 & 1 & t-3 \end{vmatrix} = (t-1)(t-2)^2.$$

So the eigenvalues are 1 and 2. The eigenspaces are

$$E_1 = \ker \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ -1 & 1 & -2 \end{pmatrix} = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \; : \; \alpha \in \mathbb{R} \right\}$$

and

$$E_2 = \ker \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ -1 & 1 & -1 \end{pmatrix} = \left\{ \alpha \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \; : \; \alpha \in \mathbb{R} \right\}.$$

Example 2

Example. Find the eigenvalues and corresponding eigenspaces of the matrix $A = \begin{pmatrix} 3 & 1 & -3 \\ 1 & 3 & -3 \\ 1 & 1 & -1 \end{pmatrix}$.

Solution. The characteristic polynomial is

$$p_A(t) = \det(tI - A) = \begin{vmatrix} t-3 & -1 & 3 \\ -1 & t-3 & 3 \\ -1 & -1 & t+1 \end{vmatrix} = (t-1)(t-2)^2.$$

So the eigenvalues are 1 and 2. The eigenspaces are

$$E_1 = \ker \begin{pmatrix} -2 & -1 & 3 \\ -1 & -2 & 3 \\ -1 & -1 & 2 \end{pmatrix} = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \ : \ \alpha \in \mathbb{R} \right\}$$

and

$$E_2 = \ker \begin{pmatrix} -1 & -1 & 3 \\ -1 & -1 & 3 \\ -1 & -1 & 3 \end{pmatrix} = \left\{ \alpha \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} \ : \ \alpha, \beta \in \mathbb{R} \right\}.$$

Characteristic polynomial for linear transformations

Lemma. The characteristic polynomial is a matrix similarity invariant.

Proof. Suppose $A, B \in M_{n,n}(\mathbb{F})$ are similar, so that $B = P^{-1}AP$ for some $P \in \mathrm{GL}(n, \mathbb{F})$. Then

$$\begin{aligned} p_B(t) = \det(tI - B) &= \det(P^{-1}(tI - A)P) \\ &= \det(P^{-1})\det(tI - A)\det(P) = \det(tI - A) = p_A(t). \end{aligned}$$

As a corollary, since transformation matrices are similar, we can generalise our definition for the characteristic polynomial for a linear transformation:

Definition. The characteristic polynomial of a vector space homomorphism $T \in \mathcal{L}(V, V)$ is given by $p_T(t) = p_A(t)$ where A is any transformation matrix of T.

Lemma. Let $T \in \mathcal{L}(V, V)$ for finite-dimensional V and let $A = [T]_B^B$ for some basis B of V. Then any $\mathbf{v} \in V$ is a λ-eigenvector of T if and only if $[\mathbf{v}]_B$ is a λ-eigenvector of A.

Proof. We have that $\lambda\mathbf{v} = T(\mathbf{v})$ if and only if $\lambda[\mathbf{v}]_B = [T(\mathbf{v})]_B = A[\mathbf{v}]_B$ by linearity of taking coordinates.

Multiplicities

Suppose $\mathbb{F}$ is a splitting field, a field for which all polynomials can be factorised into linear factors (e.g. $\mathbb{C}$). Given a matrix $A \in M_{n,n}(\mathbb{F})$ or linear map $T \in \mathcal{L}(V, V)$ with $\dim(V) = n$, we have the following definitions.

Definition. The algebraic multiplicity of an eigenvalue λ is the multiplicity of λ as a root of the characteristic polynomial, that is, the largest power of $(t - \lambda)$ that divides the characteristic polynomial.

This means that, counting algebraic multiplicities, A or T has exactly n eigenvalues.

Another multiplicity that is useful when trying to understand the behaviour of A or T is defined below.

Definition. The geometric multiplicity of an eigenvalue λ is the dimension of the λ-eigenspace, $\dim(E_\lambda)$. In the case of a linear map T this is $\operatorname{nullity}(T - \lambda \operatorname{id})$, and in the case of a matrix A this is $\operatorname{nullity}(A - \lambda I)$.

Example. The matrix from Example 1 has an eigenvalue 2 that has algebraic multiplicity 2 but geometric multiplicity 1. The matrix from Example 2 has an eigenvalue 2 that has algebraic multiplicity 2 and geometric multiplicity 2.

Properties of the characteristic polynomial

Lemma. Given $A \in M_{n,n}(\mathbb{F})$, writing its eigenvalues including algebraic multiplicities as $\lambda_1, \dots, \lambda_n$, we have $\det(A) = \prod_{i=1}^{n} \lambda_i$.

Proof. We know that $p_A(t) = \det(tI - A) = \prod_{i=1}^{n}(t - \lambda_i)$. Setting $t = 0$, we get $\det(-A) = (-1)^n \prod_{i=1}^{n} \lambda_i$, and since $\det(-A) = (-1)^n \det(A)$, the result follows.

Lemma. Given $A \in M_{n,n}(\mathbb{F})$, writing its eigenvalues including algebraic multiplicities as $\lambda_1, \dots, \lambda_n$, we have $\operatorname{tr}(A) = \sum_{i=1}^{n} \lambda_i$.

Proof. We know that

$$p_A(t) = \det(tI - A) = \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) \prod_{i=1}^{n} (tI - A)_{i,\sigma(i)}.$$

Notice that the only permutation that can select more than $n-2$ copies of t from the tI matrix is the identity permutation, since any other permutation must select at least two zeros. So the coefficient of t^{n-1} is the coefficient in $\prod_{i=1}^{n}(tI - A)_{i,i} = \prod_{i=1}^{n}(t - a_{i,i})$, which is $-\operatorname{tr}(A)$. Similarly, the coefficient of t^{n-1} in $\prod_{i=1}^{n}(t - \lambda_i)$ is $-\sum_{i=1}^{n} \lambda_i$, so $\operatorname{tr}(A) = \sum_{i=1}^{n} \lambda_i$.

Comparing multiplicities

Theorem. Given $T \in \mathcal{L}(V, V)$ for finite-dimensional V, for each eigenvalue λ of T, the geometric multiplicity of λ is less than or equal to the algebraic multiplicity of λ.

Proof. Suppose the geometric multiplicity of λ is k, so that $\dim(E_\lambda) = k \leq n$. Then we can set $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ to be a basis for E_λ. This basis can be extended to a basis $B = \{\mathbf{v}_1, \dots, \mathbf{v}_k, \dots, \mathbf{v}_n\}$ for V. Since $T(\mathbf{v}_i) = \lambda \mathbf{v}_i$ for each $1 \leq i \leq k$, we know the transformation matrix has the form

$$[T]_B^B = \begin{pmatrix} \lambda I_k & X \\ \mathbf{0} & Y \end{pmatrix}$$

for some $X \in M_{k,n-k}(\mathbb{F})$ and $Y \in M_{n-k,n-k}(\mathbb{F})$ (and $\mathbf{0} \in M_{n-k,k}(\mathbb{F})$). So

$$p_T(t) = \det(tI - [T]_B^B) = \begin{vmatrix} tI_k - \lambda I_k & -X \\ \mathbf{0} & tI_{n-k} - Y \end{vmatrix},$$

which can be partially calculated by expanding along the first k columns to give $p_T(t) = (t - \lambda)^k p_Y(t)$. This means that $(t - \lambda)$ divides the characteristic polynomial at least k times, so the algebraic multiplicity is greater than or equal to the geometric multiplicity k.

Diagonalisation

Definition. A matrix $A \in M_{n,n}(\mathbb{F})$ is called diagonal if its only nonzero entries lie on the main diagonal (the (i,i)th entries for each $1 \leq i \leq n$).

Definition. A matrix $A \in M_{n,n}(\mathbb{F})$ is diagonalisable if there is a diagonal matrix $D \in M_{n,n}(\mathbb{F})$ which is similar to A, that is, there exists $P \in \mathrm{GL}(n,\mathbb{F})$ such that $D = P^{-1}AP$.

Likewise, a mapping $T \in \mathcal{L}(V,V)$ is called diagonalisable if there is a basis B of V for which $[T]_B^B$ is diagonal.

Lemma. For finite-dimensional V, a map $T \in \mathcal{L}(V,V)$ is diagonalisable if and only if V has a basis whose elements are all eigenvectors of T.

Proof. If T is diagonalisable, then it has a diagonal transformation matrix over some basis $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, meaning $[T(\mathbf{v}_i)]_B = [\lambda_i \mathbf{v}_i]_B$ where λ_i is the (i,i)th entry of the transformation matrix. So $T(\mathbf{v}_i) = \lambda_i \mathbf{v}_i$ for each i.

If the set of T eigenvectors $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for V, then $[T(\mathbf{v}_i)]_B = [\lambda_i \mathbf{v}_i]_B = \lambda_i$ for each i where λ_i is the corresponding eigenvalue, meaning $[T]_B^B$ is the diagonal matrix with each λ_i as its diagonal entries.

Corollary. A matrix $A \in M_{n,n}(\mathbb{F})$ is diagonalisable if and only if $\mathbb{F}^n$ has a basis whose elements are all eigenvectors of A.

Diagonalisation construction

Lemma. If a matrix $A \in M_{n,n}(\mathbb{F})$ has eigenvalues $\lambda_1, \ldots, \lambda_n$ (counting algebraic multiplicities) with corresponding eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$, then A is diagonalisable if and only if the set of eigenvectors is linearly independent.

Proof. Setting $D = (\, \lambda_1\mathbf{e}_1 \;\; \lambda_2\mathbf{e}_2 \;\; \ldots \;\; \lambda_n\mathbf{e}_n \,)$ and $P = (\, \mathbf{v}_1 \;\; \mathbf{v}_2 \;\; \ldots \;\; \mathbf{v}_n \,)$ gives

$$AP = (\, A\mathbf{v}_1 \;\; A\mathbf{v}_2 \;\; \ldots \;\; A\mathbf{v}_n \,) = (\, \lambda_1\mathbf{v}_1 \;\; \lambda_2\mathbf{v}_2 \;\; \ldots \;\; \lambda_n\mathbf{v}_n \,) = PD.$$

So A is diagonalisable if and only if P is invertible, if and only if the set of eigenvectors is linearly independent.

Corollary. Given $T \in \mathcal{L}(V, V)$ with $\dim(V) = n$ has distinct eigenvalues $\lambda_1, \ldots, \lambda_k$ for some $k \leq n$, the map T is diagonalisable if and only if $\sum_{i=1}^{k} \dim(E_{\lambda_i}) = \dim(V)$.

Proof. For each λ_i, its algebraic multiplicity is the number of linearly independent λ_i-eigenvectors that must be selected, meaning its geometric multiplicity must equal its algebraic multiplicity.

Corollary. Given $T \in \mathcal{L}(V, V)$ with $\dim(V) = n$ has distinct eigenvalues $\lambda_1, \ldots, \lambda_k$ for some $k \leq n$, the map T is diagonalisable if and only if $V = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_k}$.

Proof. This immediately follows from the fact $E_\lambda \cap E_\mu = \{\mathbf{0}\}$ for all $\lambda \neq \mu$.

Diagonalisation algorithm

We now have a general algorithm for diagonalising a matrix $A \in M_{n,n}(\mathbb{F})$:

- Find the eigenvalues $\lambda_1, \ldots, \lambda_n$ (counting algebraic multiplicities) by finding the roots of the characteristic polynomial $p_A(t)$.
- For each λ_i, check that its geometric multiplicity equals its algebraic multiplicity. If not, A is not diagonalisable.
- Find a set of corresponding linearly independent eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$.
- Construct the matrices $D = \begin{pmatrix} \lambda_1\mathbf{e}_1 & \lambda_2\mathbf{e}_2 & \ldots & \lambda_n\mathbf{e}_n \end{pmatrix}$ and $P = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \ldots & \mathbf{v}_n \end{pmatrix}$. Then $D = P^{-1}AP$.

Corollary. If a matrix $A \in M_{n,n}(\mathbb{F})$ has n distinct eigenvalues, then A is diagonalisable.

Exercise. Is the matrix $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & -1 & 3 \end{pmatrix}$ diagonalisable?

Solution. We saw in Example 1 that A has eigenvalues 1 and 2. For the eigenvalue 2, the algebraic multiplicity is 2 while the geometric multiplicity is 1. This implies A is not diagonalisable, since we cannot select two linearly independent 2-eigenvectors.

Diagonalisation example

Exercise. Diagonalise the matrix $A = \begin{pmatrix} 3 & 1 & -3 \\ 1 & 3 & -3 \\ 1 & 1 & -1 \end{pmatrix}$.

Solution. We saw in Example 2 that A has eigenvalues 1 and 2 with corresponding eigenspaces

$$E_1 = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \ : \ \alpha \in \mathbb{R} \right\} \text{ and } E_2 = \left\{ \alpha \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} \ : \ \alpha, \beta \in \mathbb{R} \right\}.$$

Since each eigenvalue has equal algebraic and geometric multiplicities, A is diagonalisable. So $D = P^{-1}AP$ where

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \text{ and } P = \begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

Note that we could have constructed D and P in many other ways, but any valid construction is sufficient.